

SCIENCE PROGRAMME 2015/16

STRAND: Earth and Space Sciences

TOPIC: Our Universe

YEAR: 10 (General + Extension)

TERM: Spring Term (9 weeks)

TIME (WEEK)	STUDENT OUTCOMES	PRACTICALS	TEXT REFERENCE	ASSESSMENT/ HOMEWORK
1	Introduction Examine celestial objects, their relative size and scale, and how they fit into the cosmic scheme of things.	ScienceWeb - Lesson 1 (What's in our universe?) ScienceWeb - Lesson 2 (Size + distance in space.)		
1-2	Origin of the Universe - Describe the competing theories on the origin of the universe - Creation, Steady State, Big Bang and Big Crunch. - Identify the evidence that supports each theory. - Understand that the Big Bang Theory is considered the best theory at the moment. - Understand that our universe is considered to be expanding. - Know the current distance to the edge of the "known" universe. - Understand how the Doppler Effect gives an indication of the movement of galaxies.	Science Web - Lesson 3 The Big Bang Theory Invest. 6.5 p. 205 SQ10	SQ 10 Ch 6.3, 6.4	Research: Origin of Universe Worksheet 6.5 - Expanding Universe Worksheet 6.6 - Big Bang
3-4	The Universe - Describe the three different types of galaxies. - Understand that constellations are patterns formed by stars. - Name and locate common examples of constellations. - Appreciate the vast distances involved in space travel. - Understand the unit light year, and perform simple calculations using it. Extension - Describe how the parallax method is used to determine stellar distances.	Worksheets 10.1 + 10.3 TG Act 25: Southern Cross Invest. 6.2 p. 199 SQ10	SQ 10 Ch 6.1	Worksheet 6.1 - Observing Stars

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5-6	Stars - Describe the layers and characteristics of our Sun. - Understand the fusion process that produces energy in the Sun and relate it to Einstein's equation ($E = mc^2$). - Understand that the colour of a star indicates its surface temperature. - Understand the difference between apparent brightness and absolute brightness.	Invest. 6.3 p. 200 SQ10 Webquest: Sun Structure Invest. 6.4 p. 203 SQ10	SQ 10 Ch 6.2	
7-8	Life Cycles of Stars - Describe the life of a star from birth to death, dependent on its size. - Understand the difference between a nova and supernova. - Describe the characteristics of a white dwarf, neutron star and black hole. - Describe the three possible pathways for the life of a star. Extension - Hertzsprung-Russell diagram for life cycles.	ScienceWeb - Lesson 4 (Life + death of stars.) Webquest: Life Cycles	SQ 10 Ch 6.2	Worksheet 6.4 - Star Life Cycles
9	Observing the Universe - Describe how radio telescopes and arrays of radio telescopes are used by astronomers and astrophysicists to observe distant parts of the universe. - Explain how orbiting space telescopes are used to gather data from deep space and how they compare with Earth-based telescopes. - Appreciate that the study of the universe and the exploration of space involves teams of specialists from different branches of science, engineering and technology.	ScienceWeb - Lesson 5 (Observing the universe.)	SQ 10 Ch 6.5	Test : The Universe and Star Life Cycles

Text Science Quest 10 (SQ 10)

Resources iTunesU course - Year 10 Science
ScienceWeb (ASTA) - <http://scienceweb.asta.edu.au/years-9-10/unit3/overview/yr910-unit3-overview.html>

INTERNET SOURCES

Great pictures of the universe <http://antwrp.gsfc.nasa.gov/apod/ap060506.html>

Trinity College Astronomy <http://observatory.trinity.wa.edu.au/index.html>

(Very good links and photos.)

NASA <http://www.nasa.gov/home/>

(Very good site for teachers and students. Has terrific information on the Space Shuttle.)